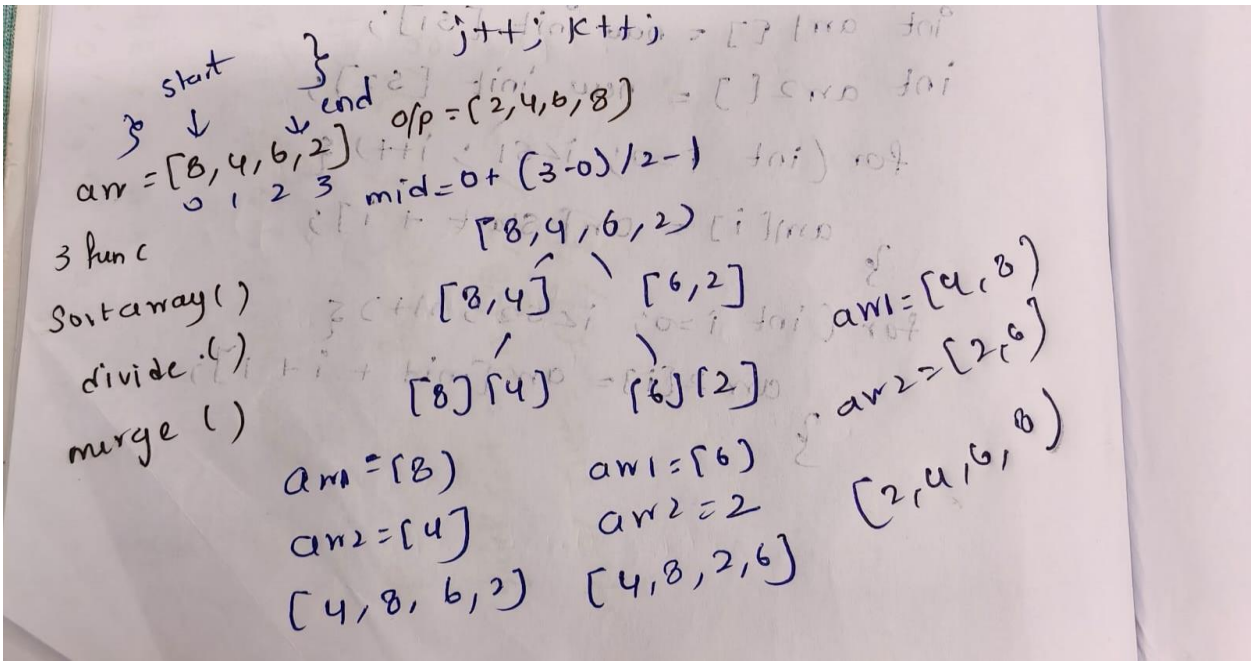
T. DHANA LAXMI
2505A41L46

1.sort an array LC-912
https://leetcode.com/problems/sort-an-array/

```java
class Solution {
        void divide(int arr[], int start, int end) {
            if(start < end) {
                int mid = start + (end - start) / 2;
               divide(arr, start, mid);
                divide(arr, mid + 1, end);
             merge(arr, start, mid, end);
            }
    }
    void merge(int arr[], int start, int mid, int end) {
            int s1 = mid - start + 1;
        int s2 = end - mid;
         int arr1[] = new int[s1];
        int arr2[] = new int[s2];
         for(int i = 0; i < s1; i++)  arr1[i] = arr[start + i];
          for(int i = 0; i < s2; i++)  arr2[i] = arr[mid + i + 1];
        int i = 0, j = 0, k = start;
         while(i < s1 && j < s2) {
         if(arr1[i] <= arr2[j]) {
                arr[k] = arr1[i];
                i++;
            }
            else {
```

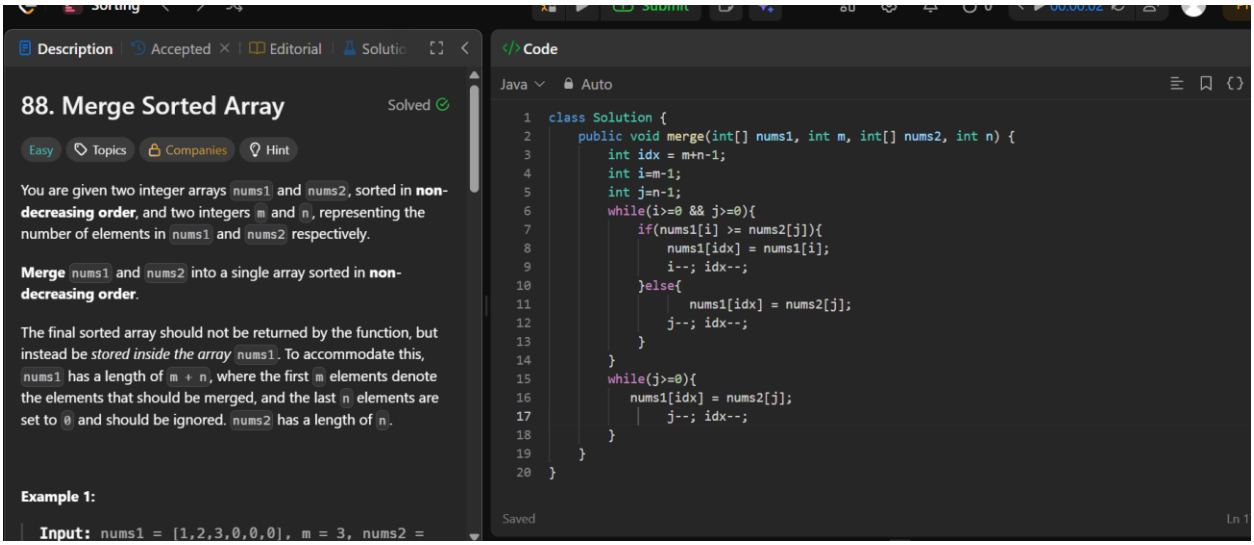
```java
                i++;
            }
            else {
                arr[k] = arr2[j];
                j++;
            }
            k++;
        }
        while(i < s1) {
            arr[k] = arr1[i];
            i++; k++;
        }
        while(j < s2) {
            arr[k] = arr2[j];
            j++; k++;
        }
    }
    public int[] sortArray(int[] nums) {
         divide(nums, 0, nums.length - 1);
            return nums;
    }
}
```

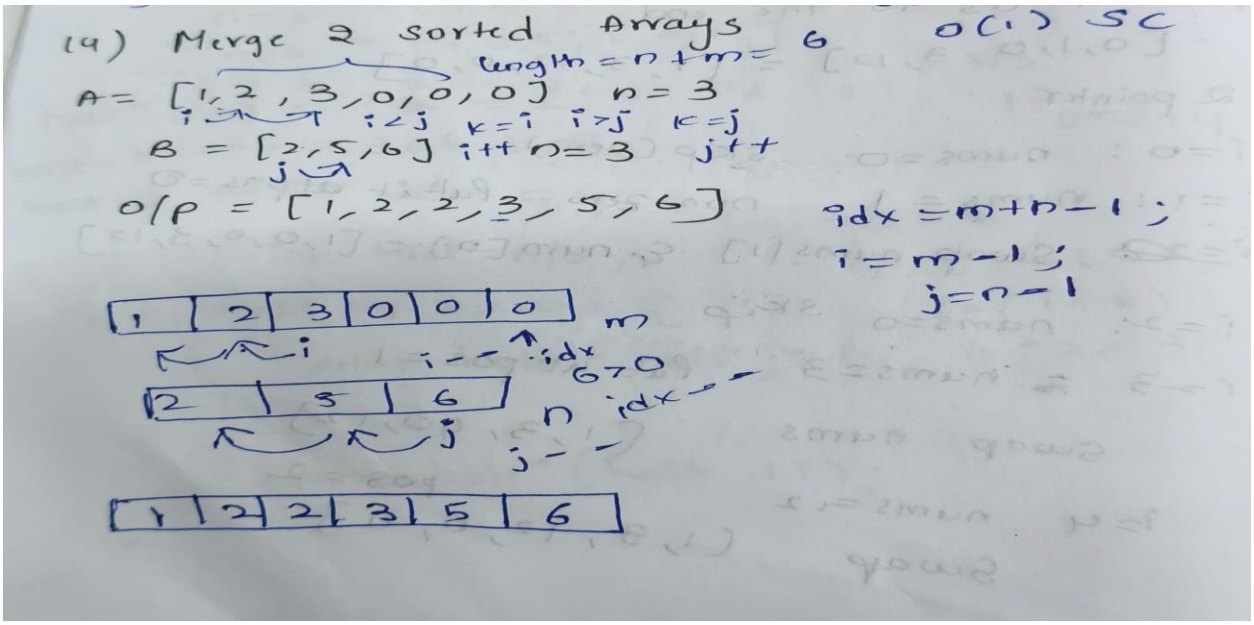
Saved



2.merge two sorted arrays

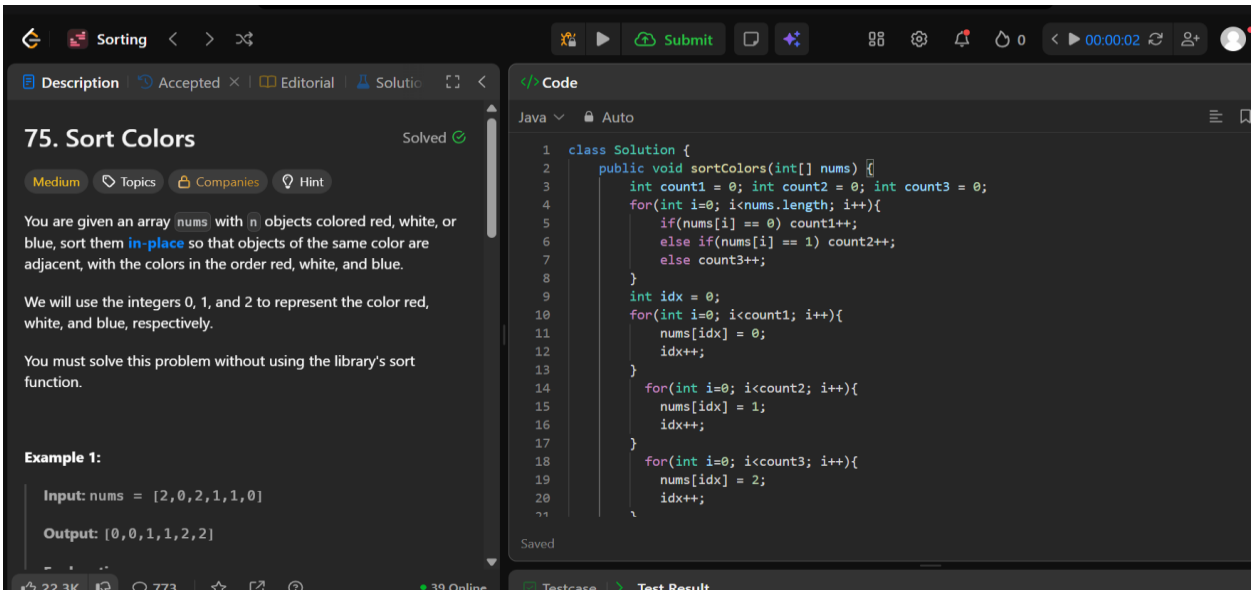
https://leetcode.com/problems/merge-sorted-array/

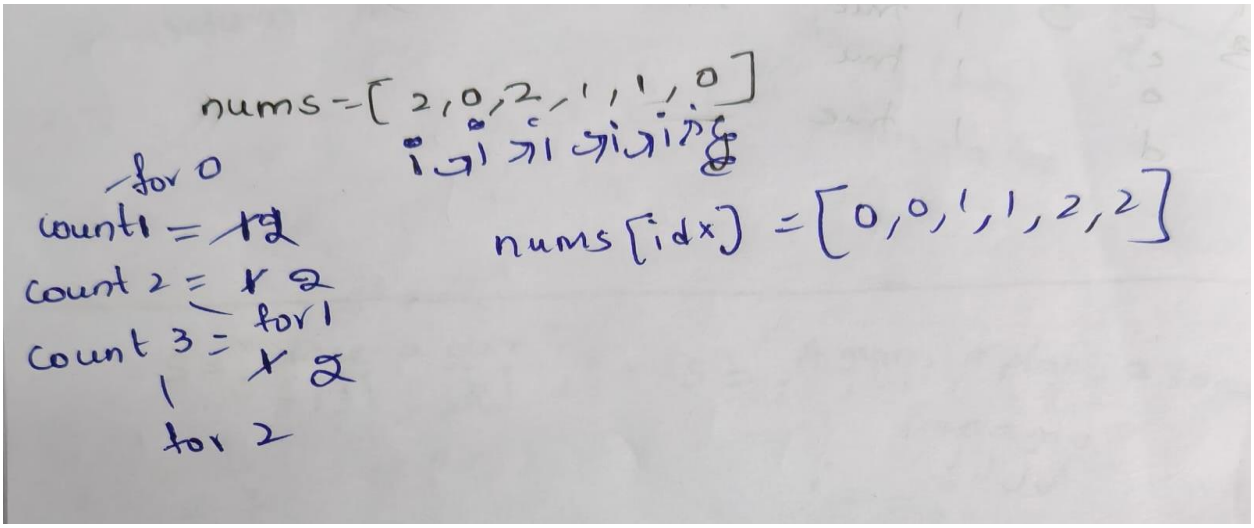




3.sort colors LC-75

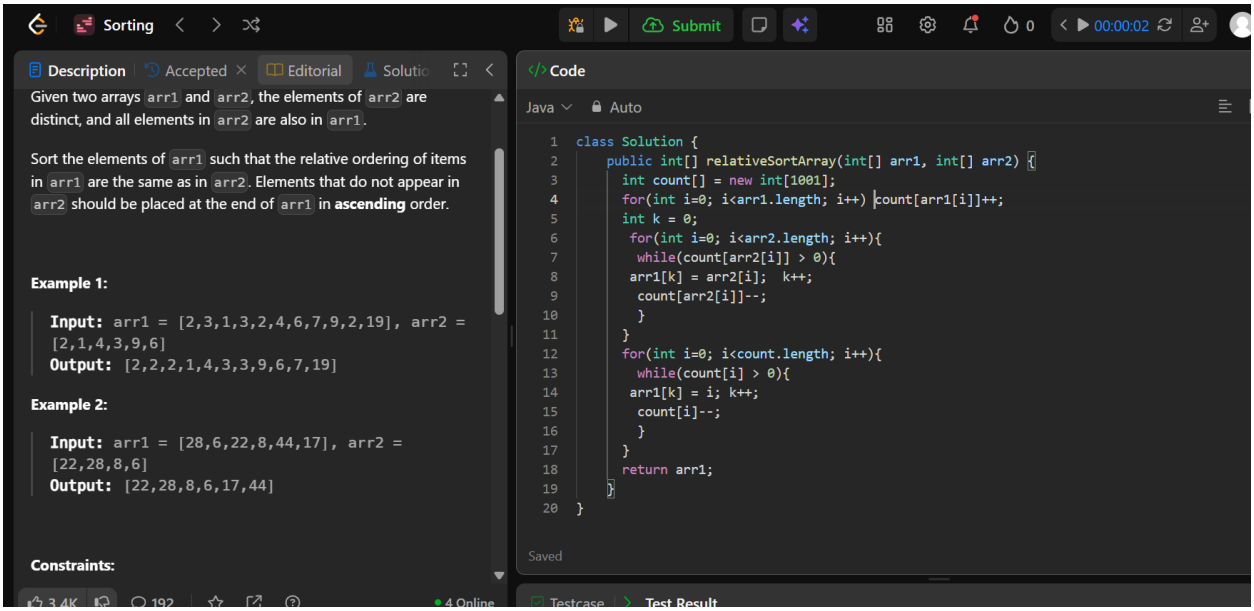
https://leetcode.com/problems/sort-colors/

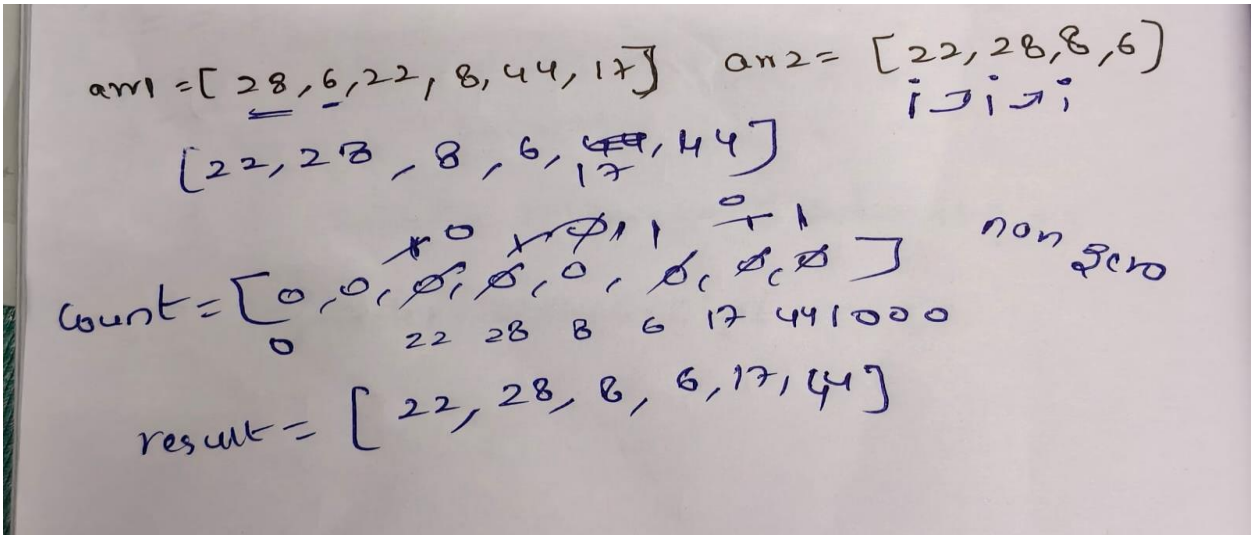




4.relative sort array LC – 1122

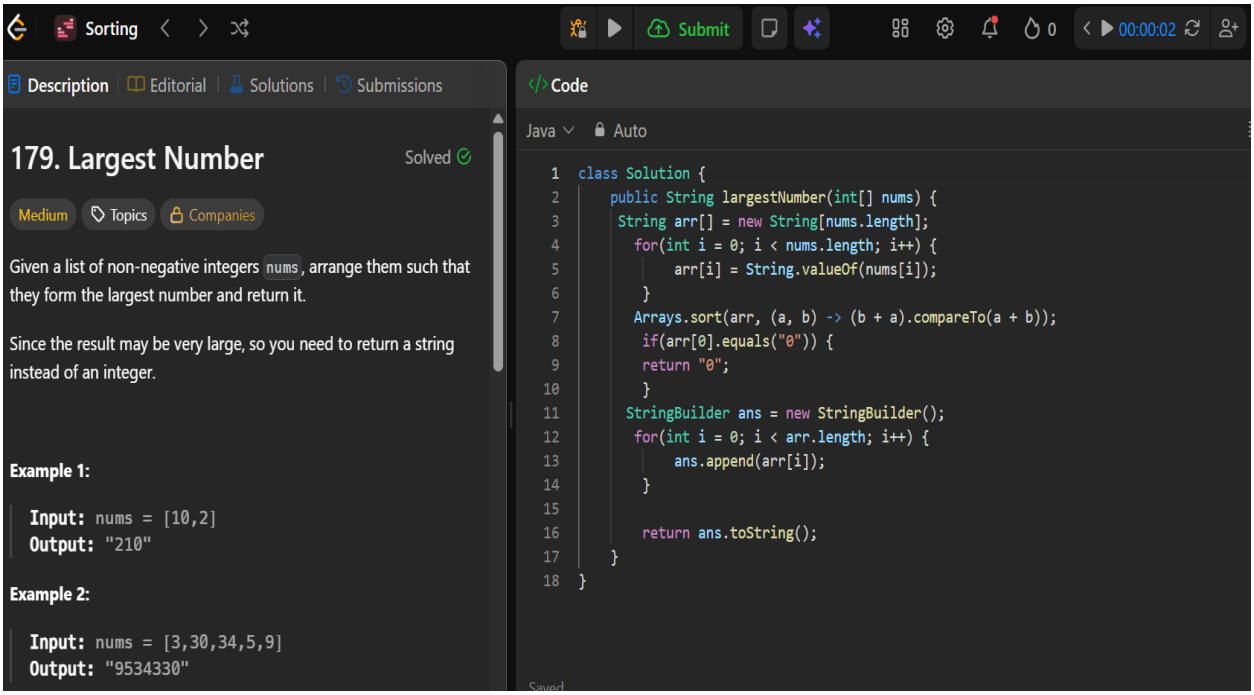
https://leetcode.com/problems/relative-sort-array/





5.largest number LC-179

https://leetcode.com/problems/largest-number/



179 largest Number

nums = [3, 30, 34, 5, 9]

↓ convert to strings → join & compare

["3", "30", "34", "5", "9"]

(a+b) → (b+a)

330 303

nums = [0, 0)

34+3 = 343

3+34 = 334

00 X

xtem 0 [9, 5, 34, 3, 30] = 9534330

a = 12

b = 121